

Total No. of Questions : 4]

SEAT No. :

PE-556

[Total No. of Pages : 2

[6578]-29

S.E. (Information Technology) (Insem.)
DATA STRUCTURE AND ALGORITHMS
(2019 Pattern) (Semester - III) (214443)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) **Answers Q. 1 or Q.2 and Q.3 or Q. 4,**
- 2) **Neat diagrams must be drawn wherever necessary.**
- 3) **Figures to the right side indicate full marks.**
- 4) **Assume Suitable data if necessary.**

- Q1)** a) Could you categorize various forms of data structures and provide an illustration for each type? **[5]**
- b) Illustrate the concepts below using examples? Data, Data Object, Data Structure. **[5]**
- c) Implement the insertion operation for adding a node at the end of a doubly linked list. Provide the algorithm and code. **[5]**

OR

- Q2)** a) Explain the concept of a linked organization in data structures. What are the advantages of using linked data structures over arrays? **[5]**
- b) What is a circular linked list, and how does it differ from a regular (singly or doubly) linked list? Give an example scenario where a circular linked list is useful. **[5]**
- c) Explain Row major and Column major representation of matrix with example? **[5]**

- Q3)** a) Analyzing the Time and Space Complexity: Provide an illustration along with a description of the time and space complexity. **[5]**
- b) What is Quick Sort, and how does it work? Explain the basic algorithm and its key steps with example. **[5]**
- c) Describe the concept of hashing in data structures. Analyse the best-case, worst- case, and average-case time complexities for hash table operations such as insertion, retrieval, and deletion. **[5]**

OR

P.T.O.

- Q4)** a) Arrange the given list in ascending order using the bubble sort algorithm. Display all the iterations List : 9, 7, -2, 4, 5, 3, -6, 2, 1, 8. [5]
- b) Can you explain the significance of the Fibonacci sequence in nature or art? Give examples. [5]
- c) For binary search, explain the best-case, worst-case, and average-case scenarios. Provide the time complexity analysis for each of these cases. [5]

